

TARUN YADAV

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WORK EXPERIENCE

SDE Intern – Synapse Technologies

Aug 2025 – Nov 2025

Bhopal

- Contributed to a real-time multiplayer gaming platform, implementing Redis-based caching and Kafka event-streaming that cut average response latency by 30% across concurrent sessions.
- Increased transaction throughput by 40% by building an asynchronous event-processing pipeline for the core game transaction engine, ensuring data consistency under concurrent load.
- Supported backend reliability during peak load by implementing distributed messaging and caching services, helping the platform maintain high availability under high-concurrency gameplay.

PROJECTS

Distributed Web Crawler – Cluster-Wide Adaptive Politeness Enforcement

[GitHub](#)

Python 3.13 (asyncio, aiohttp), Redis, SQLite, Docker, Prometheus, Hexagonal Architecture

- Designed coordinator-free domain ownership via consistent hashing (128 virtual nodes/node), resolving locally in $\sim 2\mu s$ and cutting domain reshuffling to 15.5% vs. 83% for naive modulo hashing on membership changes.
- Cut cluster-wide URL dedup memory 72x with a Redis-backed Bloom filter – 10M URLs in 17 MiB (9.6 bits/URL) with atomic check-and-add, 0.0096% false-positive rate vs. a 0.01% target.
- Replaced fixed-rate crawling with per-domain AIMD control, raising throttled-server acceptance from 0% to 94% at $\sim 1,280$ pages/sec; backed by 2,143 tests (93.4% coverage), strict mypy typing.

AlgoLab – Production-Grade Graph Algorithms Library

[GitHub](#)

Python, mypy (strict), Algorithm Design, Unit Testing

- Architected a Python graph algorithm library (20 algorithms, 10 families) processing 565k+ edges/second on 50k-node graphs – strict mypy across 65 modules, 7,000+ lines, zero type-ignore comments.
- Engineered dual max-flow implementations (Edmonds-Karp, Dinic's) delivering 40x throughput on parallel-path networks, validated by 970 tests in 1.2s with full determinism.
- Implemented Binary Lifting LCA – $O(N \log N)$ preprocessing of 65k-node trees in 253ms, $O(\log N)$ queries averaging $3.1\mu s$ – plus a Union-Find with path compression at $0.19\mu s$ amortized finds on 1M-element sets.

Barista AI – Multi-Agent Portfolio Risk Management Platform

[GitHub](#)

FastAPI, Next.js, Python, Google Gemini, NumPy/SciPy, Docker

- Orchestrated 6 specialized agents (risk analysis, market monitoring, rebalancing) sharing a singleton data-fetcher/cache layer, with Google Gemini for natural-language insights.
- Computed portfolio risk via parametric, historical, and Monte Carlo VaR (10,000 simulations), Sharpe/Sortino, and CVaR with NumPy/SciPy, backed by a rate-limited multi-source fallback chain.

EDUCATION

B.Tech in Computer Science & Engineering – Vellore Institute of Technology

Expected Sep 2026

- CGPA: 7.7/10 | Coursework: Data Structures & Algorithms, DBMS, Computer Networks, System Design, Operating Systems

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Web & APIs: React.js, Next.js, Node.js, Express.js, FastAPI, REST APIs, WebSockets

Databases: PostgreSQL, MySQL, Redis, SQLite, Kafka

Cloud & DevOps: AWS (EC2, S3, Lambda), GCP (Vertex AI), Docker, Kubernetes, CI/CD, Nginx, Git

Core CS: Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems, Computer Networks

AI/ML Tooling: LangChain, RAG Systems, Prompt Engineering, Multi-Agent Systems

CERTIFICATIONS & PROGRAMS

McKinsey Forward Program – McKinsey & Company

June 2026

Advanced Software Engineering Job Simulation – Forage (Midas)

April 2026

Cloud Computing – NPTEL

April 2024